

Semester-wise Pattern for Students Admitted to B.E. Manufacturing Programme								
Year	First Semester			U	Second Semester			U
I	BITS	F103	Introduction to Engineering Design and Prototyping	4	BITS	F111	Thermodynamics	3
	BIO	F101	Introduction to Biological Sciences	3	BITS	F112	Technical Report Writing	2
	CHEM	F101	Fundamentals of Chemistry	3	CS	F111	Computer Programming	4
	MATH	F101	Multivariable Calculus	3	MATH	F113	Probability and Statistics	3
	PHY	F101	Oscillations and Waves	3	MATH	F102	Linear Algebra and Complex Variables	3
	BITS	F101	Social Conduct	1	EEE	F111	Electrical Sciences	3
	BITS	K101	Physical Well-being and Creativity	1*	BITS	F102	Innovation and Design Thinking	1
				18				19
II	MATH	F211	Mathematics III	3	BITS	F225	Environmental Studies	3
	MF	F211	Mechanics of Solids	3	ECON	F211	Principles of Economics	3
	MF	F216	Materials Science & Engineering	3			or	or
	MF	F217	Machine Drawing	2	MGTS	F211	Principles of Management	3
	MF	F218	Transport Phenomena in Manufacturing	4	MF	F219	Operations Management	3
			Humanities Electives	3(min)	MF	F220	Metrology and Quality Assurance	3
					MF	F221	Mechanisms and Machines	3
					MF	F222	Casting, Forming and Welding	4
			18(min)				3(min)	22(min)
Summer BITS F221 Practice School – I (for PS Option Only)								
III			Open/Humanities Electives	2 to 5			Open/Humanities Electives	2 to 5
	MF	F314	Design of Machine Elements	3	MF	F317	Computer Aided Design and Manufacturing	3
	MF	F315	Automation and Control	4	MF	F318	Non Traditional Manufacturing Processes	3
	MF	F316	Machining and Machine Tools	4	MF	F319	Supply Chain Management	3
			Discipline Electives	6(min)	MF	F320	Engineering Optimization	3
			19/22				3(min)	17/20
IV			Open Electives	7 to 13	BITS	F412	Practice School-II	20
			Discipline Electives	3(min)			or	or
					BITS	F421T	Thesis	16
			10/16	or	F425T	Thesis	18-20	
				or	F424T	Thesis (9) and Electives	15-18	
							(6 to 9)	15/20

Discipline Core - 48 Units (15 Courses)
Discipline Electives - 12 Units (4 Courses)